Adam Cheng 鄭博修

R&D ASSISTANT MANAGER • CLOUD SOLUTIONS ARCHITECT • TECHNICAL PM

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EXECUTIVE PROFILE

Accomplished **R&D Assistant Manager and Cloud Solutions Architect** with **18+ years** of software engineering expertise, including **12+ years at QNAP Systems** spearheading global cloud infrastructure, core identity platforms, and enterprise administration systems. Proven track record leading multi-disciplinary engineering teams to architect, deploy, and scale high-concurrency cloud systems—including centralized Identity & Access Management (**Account Center**), multi-tenant enterprise organizational administration (**Organization Center**), microservices governance portals (**AMP**), and automated cross-cloud telemetry pipelines. Pioneer in integrating AI-augmented developer tooling (Claude Code skills & MCP), combining deep systems architecture (OAuth 2.0 / OIDC, Micro-Frontends, Fastify/Node, Python/Tornado, K8s) with formal design education from Art Center College of Design.

CORE COMPETENCIES & TECHNICAL ARCHITECTURE

Cloud & Distributed Systems: Microservices, Micro-Frontends (Module Federation), Redis Streams, Multi-Tenancy, K8s, Docker, Nginx, High Availability (99.99%), System Fault Tolerance.

Data Engineering & Telemetry: AWS (S3, Athena, CloudFront), GCP (BigQuery, GCS), PostgreSQL (async SQLAlchemy), MongoDB, Redis, Prometheus/Grafana, Celery/RabbitMQ, Looker Studio.

Identity & Security (IAM / PSIRT): OAuth 2.0 (Device Code), SAML SSO, OIDC, 2FA (TOTP/Push/SMS), WeChat Login, Enterprise RBAC/ABAC, PSIRT Remediation (BOLA, XSS, CORS, MFA fatigue).

Leadership & AI Developer Tooling: Technical Program Management (TPM), Architecture Decision Records (ADRs), Claude Code Skills & MCP (Jira, Confluence, GitLab, Grafana, MongoDB), CI/CD.

PROFESSIONAL EXPERIENCE

R&D Assistant Manager & Technical Program Manager • QNAP Systems, Inc. (威聯通科技)

Apr 2014 – Present | Taipei, Taiwan

Leading cloud backend engineering, platform architecture, security governance, and cross-functional technical delivery for core global services.

Account Center (Modern Identity Platform & Micro-Frontend Rewrite)

React 19 • Webpack 5 Module Federation • Fastify 5 (ESM) • Redis • MongoDB • K8s • Celery/RabbitMQ

- **Core Identity Gateway:** Architected QNAP's centralized identity core for myQNAPcloud ID serving millions of global accounts across separate Global and China deployments; authored ~2,030 of 2,216 commits across 6 repositories.
- **Authentication & Federation:** Implemented sign-in/up, multi-factor authentication (TOTP, SMS, Email, Push), OAuth 2.0 provider with Device Code Flow, SAML SSO, and WeChat / Mini-Program authentication flows.
- **Full-Stack Modernization:** Re-engineered legacy auth into a micro-frontend architecture using React 19 + React Compiler, React Router 7, Zustand, TanStack Query, i18next (22+ languages), and Fastify 5 Node ESM backend with multi-arch Docker and Kubernetes.
- **PSIRT Security Ownership:** Handled end-to-end security vulnerability reports from triage to production verification, successfully remediating BOLA, open redirect, DOM XSS, CORS misconfigurations, and MFA push fatigue.

Organization Center (B2B Multi-Tenant Enterprise Portal)

React • Vite • Express BFF • Python 3.14 • Tornado • Motor (async MongoDB) • Prometheus • AWS S3

- **Multi-Tenant Architecture:** Designed frontend and backend (1,350+ commits) enabling enterprise customers to govern multi-tiered organizations, members, groups, sites, invitations, webhooks, and SSO for distributed NAS hardware fleets.
- **Full-Stack Delivery:** Built Express BFF layer (Redis sessions, CSRF protection) paired with Python 3.14 Tornado async backend, GeoIP2 resolution, Chart.js analytics, and Google Maps / Baidu Maps dual geolocation visualization.

License Server & License Center

Python • Tornado • aiomysql • Motor • Redis • ECDSA / Cryptography • OpenSpec • uv

- **Entitlement & Verification:** Built backend APIs issuing and activating software licenses for QNAP NAS devices, supporting multi-seat seat-recovery, product keys, organization-restricted licensing, and China-CDN activation delivery.
- **Concurrency & Cryptography:** Engineered concurrency-safe multi-seat recovery, cryptographic signature validation (ECDSA), and resolved critical BOLA vulnerabilities. Enforced code health with uv, pytest, and ruff/mypy/bandit.

Application Management Platform (AMP) & Internal Admin Panel

Refine • React • TypeScript • Tailwind • Azure AD MSAL • Python 2.7 / 3.13 dual-runtime • Nginx

- **Core Platform Migration:** Governed the core myQNAPcloud backend (user accounts, device registration, app lifecycle), deploying Python 2.7 and Python 3.13 side-by-side behind Nginx to achieve seamless, zero-downtime progressive migration.
- **Modern Admin Hub:** Built internal administration portal using Refine, shadcn/ui, and TypeScript, featuring auto-generated CRUD pages, Microsoft Azure AD SSO (MSAL), and fine-grained role-based route/menu access control.

High-Throughput Backend Event Services & Telemetry Pipelines

Redis Streams • Pydantic • Django • AWS Athena • GCP BigQuery • Prometheus • Grafana

- **Redis Event Handler:** Built distributed event-driven service transforming Redis key-expiry/deletion events into real-time session and ZTNA record cleanup via Redis Streams, featuring active-standby forwarder with Redis distributed lock.
- **Cross-Cloud ETL Manager:** Engineered automated pipeline extracting multi-terabyte access logs from AWS Athena (S3) into Google BigQuery, feeding executive Looker Studio BI reports (Django, Celery Beat, GCS).
- **Business Metrics Pipeline:** Built aggregator extracting orders, refunds, and SSL licenses from MongoDB into Prometheus Pushgateway for Grafana monitoring. Implemented QC Webhook Mongo service for QCloud events.

DevOps Platform Engineering & AI Developer Tooling (Claude Code / MCP)

K8s Kustomize • Bun / TypeScript CLI • Astro • Drizzle ORM • Claude Code Skills • MCP Integrations

- **Infrastructure Automation:** Standardized Kubernetes Kustomize deployments across all environments; developed `k8s-scaffold` CLI (Bun, TypeScript) for automated configuration scaffolding.
- **Observability & Delivery:** Created public Service Status platform with automated health checks (Astro, React, Drizzle, PostgreSQL, Better Auth with MS SSO) and Release Helper dashboard integrated with GitLab API and container security scanning.
- **AI Engineering Ecosystem:** Pioneered Claude Code skills and Model Context Protocol (MCP) integrations across Jira, Confluence, GitLab, Grafana/Loki, and MongoDB. Built `qnap-sdd` (runs automated test suites, writes OpenSpec design docs, posts Jira reports; 259+ tests) and `qnap-psirt` vulnerability triage workflows.
- **Database Modernization (R&D):** Engineered automated migration engine transforming denormalized MongoDB collections into normalized PostgreSQL schemas with referential integrity, UUID generation, and dual-ID compatibility (async SQLAlchemy 2.0 PoC).

Software Developer • CoreLogic, Inc. [NYSE: CLGX]

2007 – Aug 2013 | Irvine / Sacramento, CA, USA

Enterprise financial & real estate data analytics for Fortune 500 mortgage and lending institutions.

- **Enterprise Core Framework:** Engineered core modules of a web-based data processing framework featuring automated job scheduling, file mapping, normalization, and ETL utilities to eliminate manual file conversion errors.
- **Quantum Data Analyzer (QDA) & Data Prep:** Developed distributed internal client-facing applications aggregating property data from hundreds of disparate county and financial sources into standardized real-time feeds (Java, Struts 2, Spring, MySQL).

Prior Engineering & Technical Consulting Highlights

2006 – 2014

- **Sarkissian Mason & Wardenclyffe Project (2011 – 2014):** Web Technology Consultant & Developer for international brands.
- **Volt Services Group (2007):** Software Developer. Engineered LPS Security Tracker (J2EE/EJB) and popDBF CLI utility.

EDUCATION & PROFESSIONAL DEVELOPMENT

Bachelor of Science (B.S.) in Engineering • Computer Science
University of Victoria (UVic), BC, Canada • 2000 – 2005

Specialized Studies in Motion & Graphic Design
Art Center College of Design (@ Night), Pasadena, CA • 2006

Graphics for Wireless & Embedded Systems
UCLA Extension, Los Angeles, CA • 2005

LANGUAGES & TECHNICAL PROFICIENCIES

Languages: English (Native / Full Bilingual Fluency), Mandarin Chinese (Native), Taiwanese (Native).

Frontend & Modern Web: React 19, Module Federation, Vite, Astro, Tailwind, shadcn/ui, Zustand, TanStack Query, i18next.

Core Languages: Python (Tornado, Django, Fastify/Node ESM), TypeScript, JavaScript, Go, Java, SQL, Bun, Bash.

Cloud, DevOps & AI: Kubernetes (Kustomize), Docker, AWS (S3, Athena, CloudFront), GCP (BigQuery), Claude Code / MCP, GitLab CI.